5 **(a)** A laser beam of red light of wavelength 644 nm is placed in front of two slits as shown in Fig. 5.1.

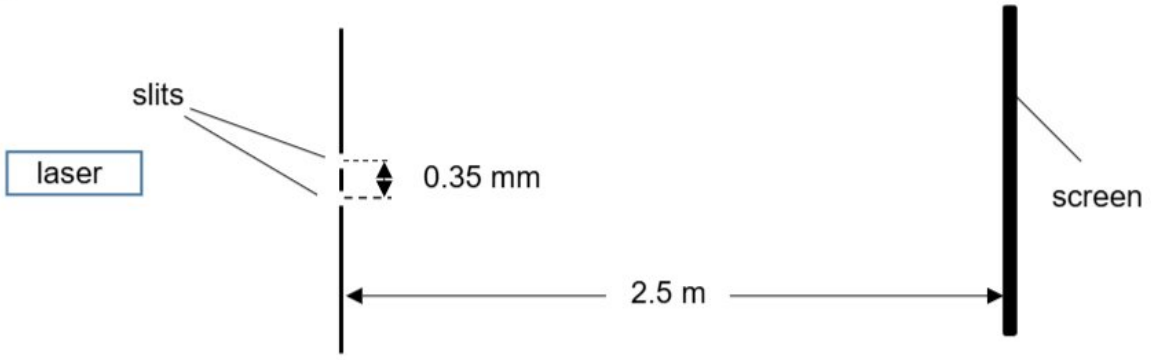


Fig. 5.1

The distance from the slits to the screen is 2.5 m. The separation of the slits is 0.35 mm. The width of each slit is 2.4×10^{-5} m.

An interference pattern consisting of dark and bright bands is formed on the screen.

(i) Each slit is now covered with a polaroid with their transmission axes aligned. One polaroid is then rotated.

Describe the effects, if any, on the fringe pattern formed on the screen as the polaroid is being rotated.

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..... [2]

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(ii) The polaroids are removed and one slit is now covered with an opaque object, as shown in Fig. 5.2.

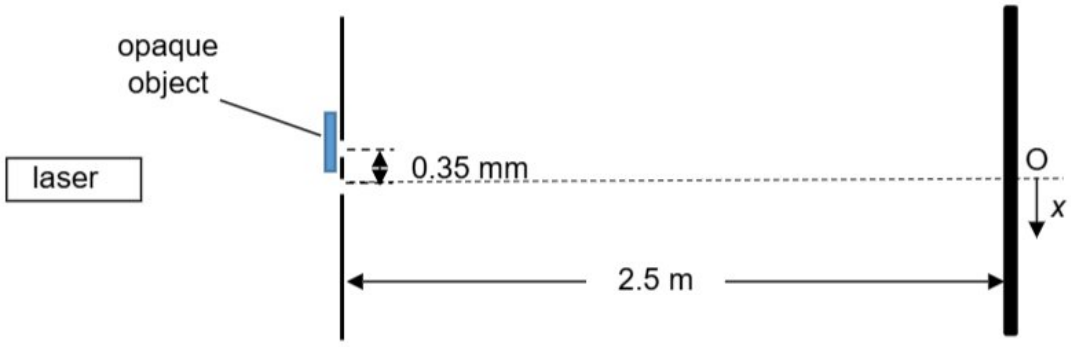


Fig. 5.2

An interference pattern is observed on the screen centred at O.

1. Calculate the width of the central fringe observed on the screen.

width = ... m [3]

2. On Fig. 5.3, sketch a graph to show the variation with distance *x* from point O of the intensity of the light observed on the screen. Draw at least 3 maxima.

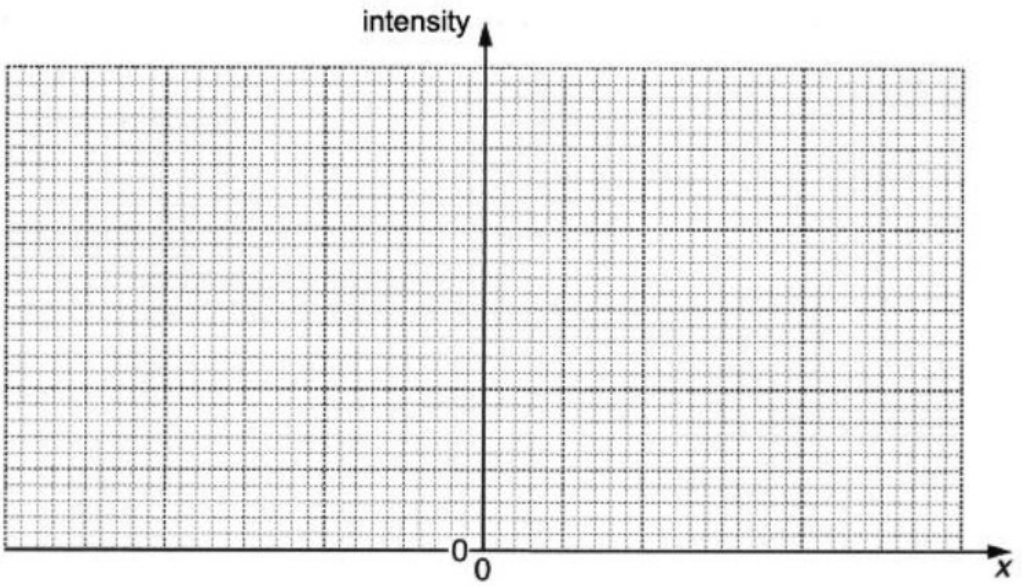


Fig. 5.3

[3]

(b) The same laser together with another identical laser are positioned in Fig. 5.4 such that the light from both sources pass through the uncovered slit at an angle of θ to each other.

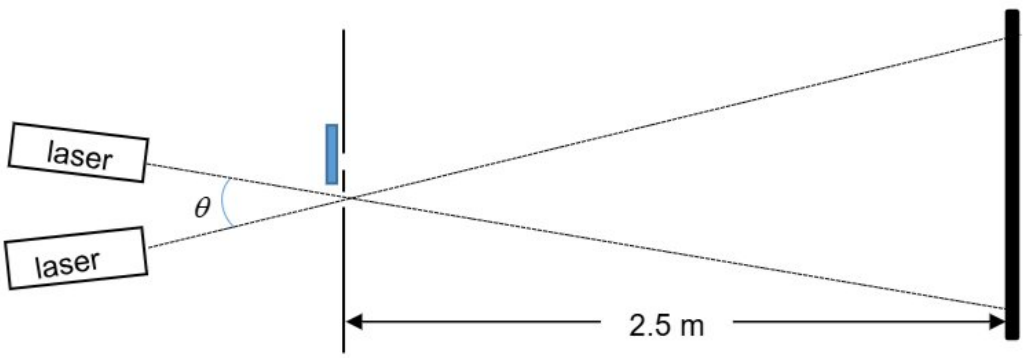


Fig. 5.4

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(i) Explain what is meant by the *Rayleigh criterion* for the resolution of the two patterns.

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..... [1]

(ii) State and explain whether the interference patterns formed by the two sources are resolved if θ is equal to 2.0°.

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..... [1]

[Total: 10]